

Mohammad Erfan Hamzei

Full-Stack Developer | Django · React · Next.js

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🌐 erfan hamzei

📍 Hamedan, Iran

👤 Professional Summary

Full-Stack Developer with **5+ years of hands-on experience** designing, building, and deploying production-grade web applications end-to-end – from responsive **React.js / Next.js** frontends to robust **Django** backends and REST APIs. Delivered **8+ successful commercial and personal projects** for businesses, cafés, and platforms, and built a proprietary **AI-assisted development framework** that speeds up client delivery from scratch to production. Comfortable owning the complete lifecycle: architecture, UI implementation, database design, API development, testing, deployment, and maintenance. Complementary background in applied **machine learning, computer vision, and deep learning** from earlier research work and personal projects.

🔧 Core Skills

Full-Stack Web Development

Django Django REST Framework React.js Next.js TypeScript JavaScript (ES6+)
Tailwind CSS HTML5 / CSS3 Bootstrap 5 REST API Design PostgreSQL / SQL
Git & GitHub Linux AI-Assisted Dev (Claude Code)
Responsive & SEO-Friendly Design Deployment & Maintenance

AI, Machine Learning & Scientific Computing

Python PyTorch TensorFlow OpenCV Computer Vision
Deep Learning C++ MATLAB

🏢 Work Experience

Freelance Full-Stack Developer

Self-Employed

Iran / Remote

2021 – Present

- **5+ years** designing and shipping production-ready websites and web applications end-to-end using **React.js, Next.js, TypeScript, Tailwind CSS, Django**, and REST APIs.
- Delivered **8+ end-to-end projects** – business sites, café brands, a job portal, and a bilingual recruitment platform – owning everything from UI to production deployment.
- Designed and maintain a reusable **AI-assisted development framework** (project rules, quality gates, reviewer agents, hooks, and a feature-module library, now on **v2.0.0**) that standardizes and speeds up client website delivery.
- Managed the complete development lifecycle: backend/API development, database integration, testing, performance optimization, deployment, and maintenance.
- Worked directly with clients and business owners to translate requirements into functional, scalable web products.

Research Experience

AI Research Assistant – Electronics Research Institute, Sharif University of Technology

Sep 2023 – Sep 2024

Contributed to machine learning and computer vision research; developed and evaluated models for real-world object detection and recognition.

Research Assistant – University of Tehran

Sep 2022 – Sep 2023

Assisted with academic research on image processing and data analysis; supported the development of AI-focused applications.

★ Featured Projects

Kariab – Persian Job Portal

Django Django REST Framework PostgreSQL

Full-featured job recruitment platform for the Iranian market. Features AI-assisted profile and job-post generation, bilingual search, a wallet/credits system, a personality-based job-matching test, dark mode, and an admin dashboard – built on scalable backend architecture designed for dynamic, real-world data.

XAU Gold Trading Suite – Desktop & Web

PySide6 Python Sentiment Analysis

A desktop application (PySide6) and companion single-file web app for algorithmic gold (XAU) trading – featuring live price feeds, SMC/RTM signal engines, interactive candlestick charting, and Persian-language news sentiment analysis.

Afkarino Website

Next.js React.js

Complete production-ready business website with a modern, responsive interface, optimized structure, and user-friendly experience. 🌐 afkarino.com

Hebemys Café Website

React.js Next.js

Modern café website focused on responsive UI/UX, business presentation, accessibility, and performance across desktop and mobile. 🌐 hebemys.com

Café Websites – Hamadan

React.js Next.js

Multiple modern, responsive café websites built for local businesses in Hamadan – menu presentation, business information, location features, and SEO-friendly architecture, with a strong focus on UI/UX and performance.

Personal Portfolio Website

HTML5 / CSS3 JavaScript Canvas

Designed and launched a personal portfolio with a dark, neon-green aesthetic featuring animated neural-network canvas backgrounds and custom interactive effects – optimized for mobile devices and search engines.

Websites Near Completion

- › **Casablanca Café (Hamadan)** – full website design and development for a café business, currently in final development and deployment.
- › **Hakim Eyewear** – business- and product-oriented website for an eyewear brand, currently being finalized for production.
- › **Nika Company** – corporate website introducing company services, business information, and communication channels.
- › **Bi Nahayat Business Website** – business-oriented site for my own growth-accelerator venture, showcasing services, branding, and customer communication features.

AI & Data Science Projects (Secondary Focus)

- › **Object Detection with YOLO** (YOLOv5/v8/v10/v11) – vehicle classification, safety-helmet detection, and smart surveillance, fine-tuned on custom datasets.
- › **Persian OCR System** (Tesseract, OpenCV) – license-plate recognition, national ID card reading, and digitization of scanned documents.
- › **Face Recognition & Emotion Detection** (FaceNet, Dlib, PF-ViT) – real-time facial-landmark alignment and emotion classification for smart authentication and live monitoring.
- › **Image Quality Enhancement** (CodeFormer, Real-ESRGAN, GFPGAN) – restoring low-resolution and blurry faces for old-photo restoration and surveillance-frame enhancement.
- › **Text-to-Speech System** (Bark, VITS) – natural, emotion-controlled Persian/English speech synthesis for voice assistants and accessibility tools.
- › **AI-Powered Telegram Super-Bot** – unified bot combining object detection, OCR, face recognition, and emotion analysis with instant feedback.
- › **Chromosome Karyotype Classification** (EfficientNet-B0, Focal Loss) – tackled a $\sim 44:1$ class-imbalance problem on a team research project, reaching $\sim 65\%$ balanced accuracy.
- › **Mini Projects** – image enhancement (Real-ESRGAN), object removal (LaMa), and small React.js web apps (to-do lists, blogs).

Education

University of Tehran

Bachelor of Science (B.Sc.) in Geomatics Engineering

Successfully completed the Bachelor's degree program and graduated in 2026.

Tehran, Iran

Sep 2022 – 2026

Languages & Interests

Language Proficiency

- › Persian: Native
- › English: Intermediate

Interests

- › Listening to Music
- › Movies & Series
- › Soccer & Volleyball
- › Research
- › Learning New Skills
- › Gym

💡 What Drives Me

I am passionate about designing and shipping modern, scalable full-stack web applications – and about building the tools and systems that make that process faster and more reliable, like my own AI-assisted development framework. I continue to expand my knowledge in artificial intelligence, computer vision, and deep learning alongside this, and I am looking for opportunities where I can combine strong software engineering with product thinking to build real-world products that matter.